Problem 18.7

Deposits of $1,000 are placed in a fund at the beginning of each year for the next 20 years. After 30 years annual payments commence and continue forever, with the first payment at the end of the 30th year. Find an expression for the amount of each payment.

Solution

Let the annual effective interest rate be i, and let X be the amount of each perpetuity payment.

The deposits of $1,000 are made at the beginning of each year for 20 years, so their present value at time 0 is

$$1000\ddot{a}_{\overline{20}|}.$$

The perpetuity payments begin at the end of year 30. This means the first payment occurs at time 30, so the perpetuity-immediate is valued one period before the first payment, at time 29. Its value at time 29 is

$$\frac{X}{i}.$$

Discounting this value back to time 0 gives

$$\frac{X}{i}v^{29}.$$

Equating values at time 0,

$$1000\ddot{a}_{\overline{20}|} = \frac{X}{i}v^{29}.$$

Solving for X,

$$X = 1000i\ddot{a}_{\overline{20}|}v^{-29}.$$

Since

$$\ddot{a}_{\overline{20}|} = (1+i)\left(\frac{1-v^{20}}{i}\right),$$

we get

$$\begin{aligned} X &= 1000i\left((1+i)\frac{1-v^{20}}{i}\right)v^{-29} \\ &= 1000(1+i)(1-v^{20})v^{-29}. \end{aligned}$$

Using $v^{-1} = 1+i$,

$$\begin{aligned} X &= 1000(1+i)^{30}\left(1-(1+i)^{-20}\right) \\ &= 1000\left((1+i)^{30}-(1+i)^{10}\right). \end{aligned}$$

Hence, an expression for the amount of each payment is

$$\boxed{X = 1000\left((1+i)^{30}-(1+i)^{10}\right)}.$$